

Affordable Real Style Transfer: Training Wavelet-Decoupled Rectified Flow on a Single RTX 3060 in Minutes

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Abstract

Because the source image is already an artwork, art-to-art style transfer is easy to overclaim. We therefore calibrate evaluation with identical-image transfer (IDT): if the unchanged source already scores highly on the requested target style, a method must beat that floor to count as real transfer. Under this calibration, low LPIPS alone is not enough. A method can preserve the source well and still move in the wrong style direction.

We trace this failure to Euclidean flow matching. In standard latent coordinates, low-frequency layout and high-frequency texture share the same basis. The flow loss then favors structure preservation and makes style-conditioned motion easy to suppress. We formalize a sufficient local condition for this regime and derive three testable consequences: removing the wavelet change should hurt the CLIP-S/LPIPS frontier, matching the routed train-time and test-time path should matter, and

repeated style injection should raise damage.

We propose WD-VF, a wavelet-decoupled vector field for rectified flow in the latent space of the Stable Diffusion v1.5 EMA VAE. A single-level Haar transform splits each latent into LL, LH, HL, and HH subbands. WD-VF predicts LL/LH/HL velocities, down-weights LL supervision, routes style queries mainly through the high-frequency bands, and applies whitening and coloring only at the endpoint.

On Distinct5-WikiArt, the 903K-parameter WD-VF model trains in 3.08 minutes and stylizes 750 outputs in 83.7 seconds including VAE decode on a single RTX 3060 (12 GB), or 111.7 ms/image end-to-end. It reaches CLIP-S = 0.7213 and LPIPS = 0.2868, exceeds SaMam by +0.1397 CLIP-S while training 141× faster, and stays close to Seedream 4.5 on the same five-style protocol. Affordable real style transfer therefore depends on the right geometry, not a large compute budget.

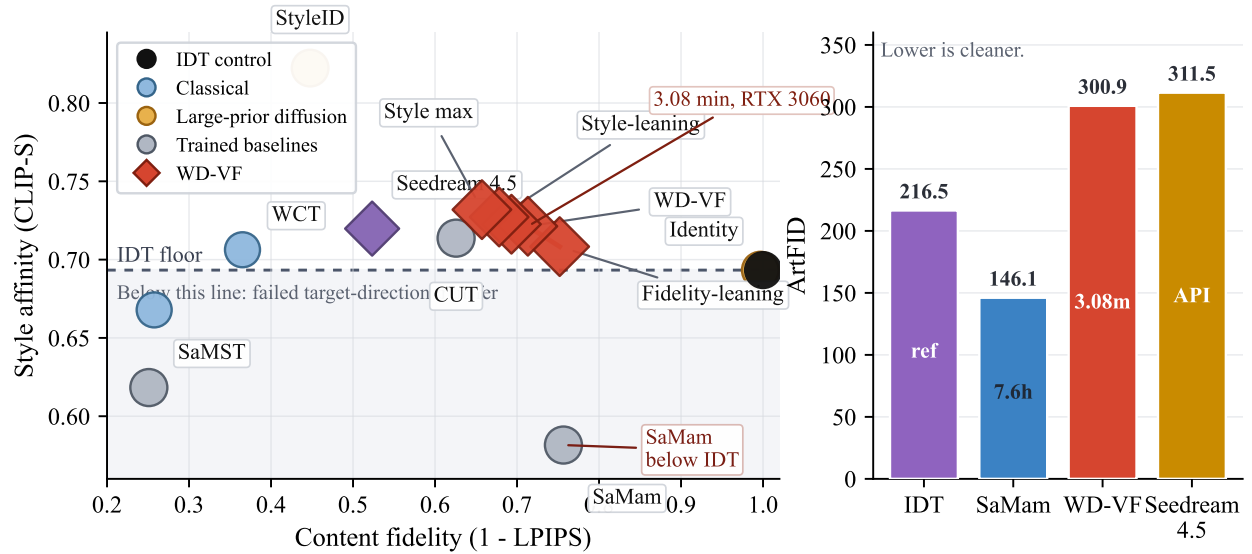


Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: all-pairs CLIP-S versus $1 - \text{LPIPS}$ under the unified 750-pair protocol. The horizontal reference is the IDT floor (0.6933). SaMam stays below that floor, so its low LPIPS reflects near-reconstruction rather than correct-direction transfer. Right: target-pooled ArtFID on the same benchmark. Numbers inside the bars report recorded training time for learned systems and interface time for external systems.

1 Introduction

Art-to-art style transfer has an unusually weak failure case: doing almost nothing can still look good in standard metrics

because the input is already an artwork. If the unchanged source already resembles the requested target style, raw target-style similarity is not evidence of transfer. We therefore measure every method against identical-image transfer (IDT): the unchanged source defines a no-op floor, and real transfer must move above that floor.

This calibration changes the ranking on Distinct5-WikiArt. We build the benchmark from the five WikiArt styles with the lowest IDT CLIP-S in a broader style pool, so the target direction is deliberately hard to fake. Under the unified 750-pair protocol, SaMam reaches LPIPS 0.2434 yet only CLIP-S 0.5816, well below the IDT floor of 0.6933. The issue is directional failure, not artifact level alone.

We trace this failure to geometry. In Euclidean latent coordinates, the rectified-flow target $u_t = z_1 - z_0$ mixes low-frequency layout motion and high-frequency texture motion in the same basis. The content term can then dominate the style term and create a local minimum in which the field suppresses style-conditioned motion. Theorem 1 formalizes a sufficient local condition for this regime. More importantly, it yields concrete predictions that we later test with ablations.

WD-VF changes coordinates instead of scaling up the generator. It applies a single-level Haar transform, down-weights LL supervision, routes style queries mainly through high-frequency bands, and stylizes only the endpoint subbands. In the ideal routed limit, the supervised residual energy decomposes by high-frequency subband. This gives a simple account of why the model trains quickly and why its ablations follow a predictable pattern.

The resulting operating point is small and explicit: 903K parameters, 3.08 minutes of training on a single RTX 3060, and 83.7 seconds to generate and decode 750 outputs (111.7 ms/image end-to-end). WD-VF reaches CLIP-S 0.7213 and LPIPS 0.2868, clears the IDT floor, trains $141\times$ faster than SaMam, and lands near Seedream 4.5 on the same five-style protocol.

Contributions. (1) We introduce IDT-calibrated evaluation for art-to-art transfer and show that it changes which baselines count as real target-direction transfer. (2) We give a local collapse analysis for Euclidean latent flow and state a sufficient curvature condition under which style-conditioned motion is suppressed (Theorem 1). (3) We propose **WD-VF**, a wavelet-decoupled rectified flow with LL down-weighting, high-frequency query routing, and endpoint subband alignment, and we show that its ideal routed limit removes the dominant low-frequency conflict from both supervision and style querying (Theorem 2). (4) We show that this geometry yields an affordable operating point: 903K parameters, 3.08 minutes of training, 83.7 seconds for end-to-end generation plus decode over 750 outputs, and the strongest positive-IDT CLIP-S/LPIPS tradeoff among the trained domain-conditional baselines in the current comparison.

2 Related Work

Exemplar-conditioned transfer. Classical exemplar-based methods (Gatys, Ecker, and Bethge 2015, 2016; Huang and Belongie 2017; Li et al. 2017; Deng et al. 2022; Huang et al. 2024) receive a target image at inference time

and transfer its statistics or correspondences. Our setting is stricter: the model sees only the source image and a target style identity. The problem is therefore domain-conditional translation rather than exemplar stylization.

Domain-conditional baselines and large systems. The closest learned baselines are compact multi-style systems such as SaMST and SaMam (Liu et al. 2024a,b). Larger diffusion-based or commercial systems such as StyleID, SDEdit, and Seedream define the high-compute end of the comparison surface (Meng et al. 2021; Chung, Hyun, and Heo 2024; Zhang et al. 2024). WD-VF targets low-cost local training under the same domain-conditional interface.

Wavelets and evaluation. Prior wavelet methods mainly use multi-scale decompositions as feature-processing modules (Daubechies 1992; Phung, Ghosh, and Tran 2023). We use a wavelet basis instead as a coordinate change that separates low-frequency structure from high-frequency style motion. We pair this design with IDT calibration because art-to-art transfer already starts inside the art manifold. The main metrics are therefore CLIP-S, LPIPS (Zhang et al. 2018), and the gap to the IDT floor. ArtFID (Wright and Ommer 2022) is reported as auxiliary evidence rather than the main success criterion.

3 Method: Wavelet-Decoupled Rectified Flow

3.1 Setup

Let $x_c \in \mathbb{R}^{H \times W \times 3}$ be a content image and $s \in \{1, \dots, K\}$ a target style domain. We work in the latent space of the Stable Diffusion v1.5 EMA VAE and write

$$z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times 32 \times 32}. \quad (1)$$

During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment

$$\begin{aligned} z_t &= (1-t)z_0 + tz_1, \\ u_t &= z_1 - z_0, \quad t \sim \mathcal{U}([0, 1]). \end{aligned} \quad (2)$$

At inference we integrate an 8-step Heun solver,

$$\begin{aligned} v_t &= v_\theta(z_t, t, s), \\ v'_t &= v_\theta(z_t + \Delta t v_t, t + \Delta t, s), \\ z_{t+\Delta t} &= z_t + \frac{\Delta t}{2} (v_t + v'_t), \end{aligned} \quad (3)$$

then decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$. Heun is the best measured cost-quality tradeoff in our ablations: Euler loses style, while RK4 adds cost without improving the final result.

3.2 Why Euclidean Latents Collapse

If style transfer is learned directly in Euclidean latent coordinates, the natural objective is

$$\mathcal{L}_{\text{euc}} = w_f \mathcal{L}_{\text{FM}} + w_s \mathcal{L}_{\text{sty}}, \quad (4)$$

where $\mathcal{L}_{\text{FM}} = \mathbb{E}_t \|\tilde{v}_\theta(z_t, t, s) - u_t\|_2^2$ and $\mathcal{L}_{\text{sty}}$ is any endpoint style-matching term. The difficulty is structural: the straight target $u_t = z_1 - z_0$ mixes low-frequency layout motion and high-frequency texture motion in the same coordinates.

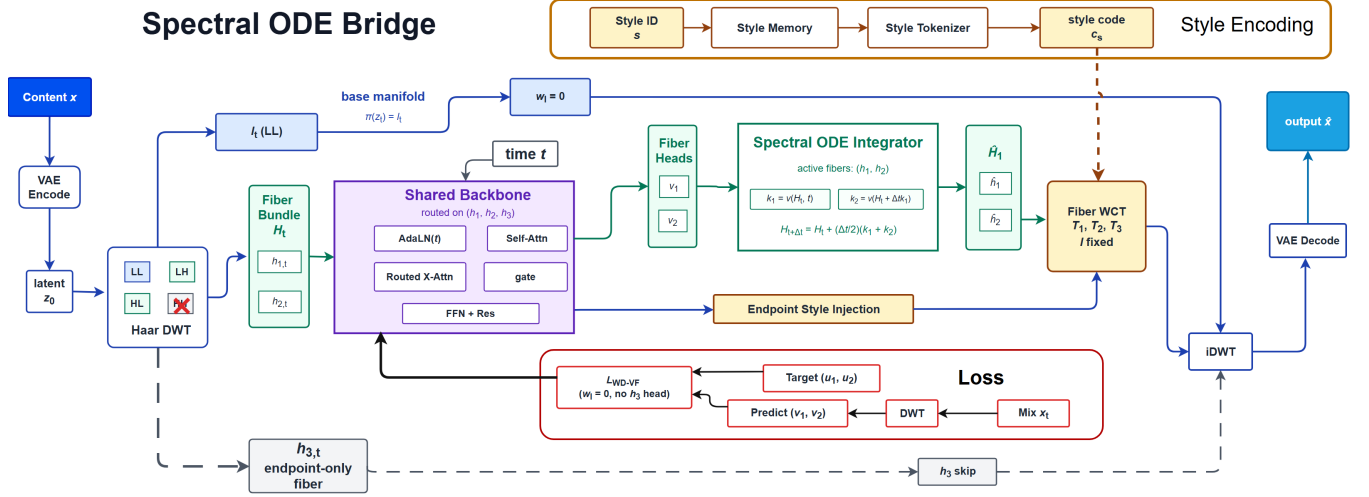


Figure 2: WD-VF transforms each latent into LL, LH, HL, and HH subbands. The diagram emphasizes the high-frequency-only query case, in which LL is removed from the style-query path. The implemented model predicts LL/LH/HL velocities, trains them with asymmetric weights (0.3, 1, 1), and applies endpoint WCT to LH/HL/HH while leaving LL untouched there.

Definition 1. Define

$$\mathcal{M}_0 = \{\theta : \tilde{v}_\theta(z, t, s) = \bar{v}_\theta(z, t) \text{ for all } z, t, s\}, \quad (5)$$

with $\bar{v}_\theta(z, t) = \mathbb{E}_s[\tilde{v}_\theta(z, t, s)]$.

The set $\mathcal{M}_0$ contains parameters whose field ignores the style label. This is the low-LPIPS, sub-IDT regime we observe empirically. The Euclidean baseline is well described by

$$\begin{aligned} \tilde{v}_\theta(z, t, s) &= \bar{v}_\theta(z, t) + \kappa(\theta) \Delta v_\theta(z, t, s), \\ \mathbb{E}_s[\Delta v_\theta(z, t, s)] &= 0, \end{aligned} \quad (6)$$

where $\kappa(\theta)$ is the retained style amplitude. Points on $\mathcal{M}_0$ satisfy $\kappa(\theta) = 0$.

Theorem 1. Let $\theta_0 \in \mathcal{M}_0$ be stationary in the normal directions:

$$\nabla_\perp \mathcal{L}_{\text{FM}}(\theta_0) = 0, \quad \nabla_\perp \mathcal{L}_{\text{sty}}(\theta_0) = 0. \quad (7)$$

Suppose the normal curvatures satisfy

$$\nabla_\perp^2 \mathcal{L}_{\text{FM}}(\theta_0) \succeq \mu I, \quad \|\nabla_\perp^2 \mathcal{L}_{\text{sty}}(\theta_0)\|_{\text{op}} \leq \nu, \quad (8)$$

If $w_f \mu > w_s \nu$, then θ_0 is a strict local minimum of $\mathcal{L}_{\text{euc}}$ restricted to the normal space $T_{\theta_0}^\perp \mathcal{M}_0$.

Proof. For any normal perturbation $\delta \in T_{\theta_0}^\perp \mathcal{M}_0$, the stationarity conditions (7) remove all first-order terms. A second-order expansion gives

$$\begin{aligned} \Delta \mathcal{L}_{\text{euc}}(\delta) &= \frac{1}{2} \delta^\top \left(w_f \nabla_\perp^2 \mathcal{L}_{\text{FM}}(\theta_0) + w_s \nabla_\perp^2 \mathcal{L}_{\text{sty}}(\theta_0) \right) \delta \\ &\quad + o(\|\delta\|^2). \end{aligned} \quad (9)$$

Using (8),

$$\Delta \mathcal{L}_{\text{euc}}(\delta) \geq \frac{1}{2} (w_f \mu - w_s \nu) \|\delta\|^2 + o(\|\delta\|^2). \quad (10)$$

The coefficient of $\|\delta\|^2$ is positive, so $\Delta \mathcal{L}_{\text{euc}}(\delta) > 0$ for all sufficiently small nonzero δ .

The two curvature bounds have a simple meaning. The flow term is locally steep in the normal directions, while the style term is locally flatter. When $w_f \mu > w_s \nu$, the cheaper local choice is to stay near $\mathcal{M}_0$ instead of following the style-conditioned direction.

Theorem 1 is explanatory, but it is not empty. It yields a concrete prediction: if the style field is learned in plain Euclidean coordinates, then removing the wavelet change should move the model toward a weaker CLIP-S/LPIPS operating point. The No-DWT row in Table 3 and the small fitted style amplitude of the Euclidean baseline ($\kappa \approx 0.016$) test exactly that prediction.

3.3 Wavelet Coordinates

WD-VF first changes coordinates. For a 2×2 patch (a, b, c, d) , the single-level Haar transform writes

$$\mathcal{W}(z) = (\ell, h_1, h_2, h_3), \quad (11)$$

with

$$\begin{aligned} \ell &= \frac{1}{2}(a + b + c + d), & h_1 &= \frac{1}{2}(a + b - c - d), \\ h_2 &= \frac{1}{2}(a - b + c - d), & h_3 &= \frac{1}{2}(a - b - c + d). \end{aligned} \quad (12)$$

Here ℓ is the low-frequency coordinate and $h = (h_1, h_2, h_3)$ collects the high-frequency coordinates.

Lemma 1. The transform is orthogonal, so

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \quad (13)$$

Definition 2. We use ℓ as the low-frequency coordinate and (h_1, h_2, h_3) as the high-frequency coordinates. Equivalently, the projection map is $\pi(z) = \ell$.

Proposition 1. Let $z = z_{\text{cnt}} + z_{\text{sty}}$ and write $\mathcal{W}(z_{\text{cnt}}) = (\ell_{\text{cnt}}, h_{\text{cnt}})$. If

$$\|h_{\text{cnt}}\|_F^2 \leq \epsilon \|z_{\text{cnt}}\|_F^2 \quad \text{for some } \epsilon \in [0, 1], \quad (14)$$

then any high-frequency-only map $T_\ell(\ell, h) = (\ell, \phi(h))$ satisfies

$$|\|T_\ell(z) - z\|_F^2 - \|\phi(h) - h\|_F^2| \leq \epsilon \|z_{\text{cnt}}\|_F^2. \quad (15)$$

Proof. Lemma 1 removes all cross terms between ℓ and h . The only content perturbation left is the small content residue already present in h .

This proposition states the geometric premise of WD-VF: when most content energy lies in LL, moving only the high-frequency coordinates changes texture much more than layout.

3.4 Training Objective

Let

$$\begin{aligned} \mathcal{W}(z_t) &= (\ell_t, h_{1,t}, h_{2,t}, h_{3,t}), \\ \mathcal{W}(u_t) &= (u_\ell, u_1, u_2, u_3). \end{aligned} \quad (16)$$

The final WD-VF model receives $(\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and predicts three velocity heads (v_ℓ, v_1, v_2) . The h_3 band is kept in the state and stylized at the endpoint, but it has no learned velocity head because the measured gain of an HH head is within ± 0.0001 CLIP-S.

The implemented training objective is

$$\mathcal{L}_{\text{impl}} = \mathbb{E}_t [\lambda_\ell \|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2]. \quad (17)$$

with

$$\lambda_\ell = 0.3. \quad (18)$$

WD-VF de-weights LL rather than eliminating it. Some styles also shift coarse color tone, so LL remains as a weak anchor term. Its weight stays strictly below the two learned high-frequency terms.

Loss-level de-weighting is only half of the construction. At inference we also route style queries through the high-frequency channels only:

$$\begin{aligned} q_{\text{routed}} &= Q[h_{1,t}; h_{2,t}; h_{3,t}], \\ k &= K(m_s), \\ v &= V(m_s), \end{aligned} \quad (19)$$

where m_s is the style memory. During training, we use the high-frequency query with probability $p = 0.8$ and the full-latent query $q_{\text{full}} = Q[\ell_t; h_{1,t}; h_{2,t}; h_{3,t}]$ with probability 0.2. This stochastic routing matches the inference geometry while preserving a small low-frequency color channel.

4 Why WD-VF Trains Quickly

4.1 High-Frequency-Only Query Case

The next theorem isolates the mechanism behind WD-VF. In this case, $\lambda_\ell = 0$ and the high-frequency query (19) is used deterministically.

Theorem 2. *In the high-frequency-only query case, the supervised WD-VF objective reduces to*

$$\mathcal{L}_{\text{ideal}} = \mathbb{E}_t [\|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2]. \quad (20)$$

That is, the learned style transport lives on two orthogonal supervised subbands: h_1 and h_2 .

Proof. Lemma 1 makes the residual norm split by subband. Setting $\lambda_\ell = 0$ removes the LL residual, and the implementation has no learned velocity head for h_3 . In the high-frequency-only query case, ℓ is also absent from the style-query path. The only supervised transport terms that remain are therefore the h_1 and h_2 residuals.

Corollary 1. *Under Theorem 2, the dominant low-frequency conflict identified by Theorem 1 is absent from both the supervised objective and the style-query path.*

This result also yields concrete predictions. First, train-time routing should matter because the best inference path is the high-frequency path. Second, repeated style injection should be unnecessary once high-frequency transport is learned cleanly, so endpoint-only alignment should preserve LPIPS better than per-step injection. We test both predictions in Tables 3 and 5.

Final model. The final WD-VF model keeps the same geometry with $\lambda_\ell = 0.3$, a high-frequency query used with probability $p = 0.8$, a full-latent query used with probability 0.2, and no HH velocity head. Inference returns to the high-frequency query, and endpoint WCT stylizes LH/HL/HH while keeping LL fixed. This setting retains a low-frequency tone channel without reopening the dominant Euclidean conflict.

4.2 Endpoint Subband Alignment

Let the integrated endpoint satisfy

$$\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3), \quad (21)$$

and let a reference style latent satisfy

$$\mathcal{W}(z_1^*) = (\ell^*, h_1^*, h_2^*, h_3^*). \quad (22)$$

For $i \in \{1, 2, 3\}$, define the whitening-coloring map

$$T_i(a) = \Sigma_i^{*1/2} \hat{\Sigma}_i^{-1/2} (a - \hat{\mu}_i) + \mu_i^*, \quad (23)$$

where $(\hat{\mu}_i, \hat{\Sigma}_i)$ are the mean and covariance of $\hat{h}_i$ and (μ_i^*, Σ_i^*) are those of h_i^* .

We then set

$$\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, T_1(\hat{h}_1), T_2(\hat{h}_2), T_3(\hat{h}_3)). \quad (24)$$

Proposition 2. *The endpoint WCT leaves the base coordinate fixed:*

$$\pi(\hat{z}_1^+) = \pi(\hat{z}_1) = \hat{\ell}. \quad (25)$$

Proof. Each map T_i acts only on h_i . The first coordinate of $\mathcal{W}(\hat{z}_1^+)$ is therefore still $\hat{\ell}$, and the inverse wavelet transform preserves that base coordinate.

WCT is used here as a terminal second-order alignment map. In the final model, the endpoint scales are (0.0, 0.3, 0.3, 0.5) for LL/LH/HL/HH respectively, so the low-frequency coordinate is left untouched while the three high-frequency bands receive progressively stronger moment matching.

We stylize only once at the endpoint to preserve strength control. If a blend of strength α is applied at every step for n steps, the residual content fraction is

$$r_n(\alpha) = (1 - \alpha)^n. \quad (26)$$

Proposition 3. For $n = 8$, even a moderate blend gives $r_8(0.5) = 3.9 \times 10^{-3}$, so repeated injection quickly saturates. Endpoint-only injection corresponds to $n = 1$, for which

$$r_1(\alpha) = 1 - \alpha \quad (27)$$

is affine and therefore preserves blend identifiability.

Proof. Each injection preserves a fraction $1 - \alpha$ of the current content residue, so composing n identical injections gives $r_n(\alpha) = (1 - \alpha)^n$. The case $n = 1$ is affine in α , whereas $n \gg 1$ compresses the same control range exponentially.

4.3 Architecture and Measured Cost

The selected checkpoint uses four residual blocks at width 64, three 3×3 velocity heads (LL/LH/HL), and 903K trainable parameters. Training runs for 5 epochs with batch size 24 and AdamW at 2×10^{-4} . The model trains end-to-end in 3.08 minutes on a single RTX 3060 (12 GB). On the aligned 750-pair packet, latent generation takes 42.9 seconds and VAE decode takes 40.8 seconds, so stylization itself takes 83.7 seconds, or 111.7 ms/image. Adding CLIP and LPIPS metrics brings the full evaluation packet to 97.3 seconds. The distinction matters because the paper claims affordable transfer, not merely affordable offline evaluation.

5 Experiments

5.1 Setup and Protocol

Dataset. We use the Distinct5-WikiArt-512 benchmark: five artistic domains — Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e. These five styles were selected from a broader WikiArt pool as the styles with the lowest IDT CLIP-S, so the benchmark intentionally emphasizes transfers that are hard to fake with no-op behavior. Each domain has 3,600 training images and 150 test images. Inputs are $4 \times 32 \times 32$ latents from the Stable Diffusion v1.5 EMA VAE for 512×512 images. Training uses 18,000 latents; evaluation uses 750 pairs (5 source domains $\times$ 5 target domains $\times$ 30 source images).

Metrics. CLIP-S is the cosine similarity between CLIP ViT-B/32 embeddings of the output and a target-style prototype (higher is better). LPIPS is the AlexNet perceptual distance between output and content image (lower is better). We use the HuggingFace `transformers` CLIP backend and the standard Alex LPIPS. All baselines are evaluated under this unified protocol.

IDT calibration. The 750-pair protocol includes 150 identity pairs (source = target domain). This is the key calibration: a method that performs *no* style transfer (Identity) achieves CLIP-S=0.6933. Any method whose CLIP-S falls below 0.6933 is staying closer to the source style than to the requested target style, even if LPIPS is low. This protocol exposes SaMam’s failure mode (CLIP-S 0.5816 < 0.6933), which LPIPS-only reporting hides.

Baselines. Twelve methods: Identity (no transfer), AdaIN, WCT (VGG19), SD-Turbo, SDEdit at $s \in \{0.35, 0.40\}$, StyleID, CUT, SaMST, SaMam, Seedream 4.5 (commercial API), and our method. SaMST and SaMam have explicit convergence packets under the aligned Distinct5 setup. CUT uses a retained Related_Works checkpoint with a recorded 322.6-minute training time but no surviving training log, so we treat its quality numbers as reliable and its optimization trace as archival.

5.2 Main Results

Table 1, Table 2, and Figure 1 support four claims.

IDT calibration changes the ranking. The identity row is not bookkeeping. It is the no-op floor for real transfer. SaMST and SaMam both fall below it. Their low LPIPS values therefore quantify source preservation, not movement toward the requested target style. This is the central evaluation correction of the paper.

WD-VF is the efficient trained-local point. Among methods retrained on Distinct5, WD-VF is the only row that is simultaneously positive-IDT, sub-0.30 LPIPS, and minute-scale to train. CUT clears the floor but trains for hours and still lands at higher LPIPS. SaMST and SaMam miss the target direction altogether. The main claim is therefore not a marginal metric gain but a different efficiency-fidelity operating point.

Large priors still define the raw CLIP-S ceiling. StyleID and SDEdit remain stronger if CLIP-S alone is the objective. WD-VF does not target that regime. It instead occupies the high-fidelity frontier: it stays numerically close to Seedream 4.5 in CLIP-S while cutting LPIPS from 0.4767 to 0.2868, a 39.8% reduction, with a 903K local model.

Cost columns are part of the result. The paper separates stylization from evaluation. The actual stylization path is 42.9 seconds for latent generation plus 40.8 seconds for VAE decode over 750 outputs, or 83.7 seconds total and 111.7 ms/image end-to-end. The 97.3-second figure is the full evaluation packet after adding CLIP and LPIPS. The scientific point is not the GPU name itself. The point is that the proposed geometry reaches a useful operating point without a large model or a long training run.

Auxiliary checks keep the same reading. Figure 1(right) adds a target-pooled ArtFID check. WD-VF reaches 300.9 versus 311.5 for Seedream 4.5, so the method stays in the same auxiliary realism range without changing the main CLIP-S/LPIPS conclusion. On the transfer-only subset, WD-VF remains positive relative to the corresponding off-diagonal IDT reference, reaching 0.6908 CLIP-S and 0.2965 LPIPS.

5.3 Ablations

Table 3 tests the design choices implied by the method analysis: the wavelet change, LL handling, endpoint alignment, routing probability, solver order, and model capacity. Each row removes or perturbs one component while keeping the rest fixed.

Table 1: Main comparison on the unified 750-pair Distinct5-WikiArt-512 protocol (5 source domains $\times$ 5 target domains $\times$ 30 source images, including 150 identity pairs). Δ_{IDT} is computed against the all-pairs identity floor 0.6933. Train reports audited training wall time when available. Infer reports recorded 750-output stylization time; for WD-VF this is generation plus VAE decode only (83.7 seconds), excluding metric computation. ‘m’ denotes minutes and ‘s’ denotes seconds. ‘-’ means that the current packet does not contain a verified number.

#	Method	Interface	CLIP-S $\uparrow$	LPIPS $\downarrow$	Δ_{IDT} $\uparrow$	Params	Train	Infer / 750
1	Identity	baseline	0.6933	0.0000	0.0000	—	—	—
2	AdaIN	classical-inf	0.6679	0.7425	-0.0253	—	—	—
3	WCT (VGG19)	classical-inf	0.7063	0.6348	+0.0130	—	—	90 s
4	SD-Turbo	diffusion-inf	0.6933	0.0033	+0.0000	—	—	—
5	SDEdit $s=0.35$	diffusion-sweep	0.7797	0.4508	+0.0864	—	—	—
6	SDEdit $s=0.40$	diffusion-sweep	0.7934	0.4826	+0.1001	—	—	—
7	StyleID	diffusion-inf	0.8223	0.5523	+0.1291	—	—	—
8	CUT	gan-train	0.7137	0.3743	+0.0204	7.0M	322.6 m	5.0 m
9	SaMST	mamba-train	0.6183	0.7490	-0.0749	8.3M	39.5 m	10.0 m
10	SaMam	mamba-train	0.5816	0.2434	-0.1116	—	436.0 m	17.6 m
11	Seedream 4.5	commercial-api	0.7198	0.4767	+0.0265	—	—	—
12	WD-VF (Ours)	spectral-flow	0.7213	0.2868	+0.0280	903K	3.08 m	83.7 s

Table 2: Timing breakdown for the selected WD-VF packet on a single RTX 3060. We separate stylization from metric computation to avoid conflating inference with evaluation.

Stage	Seconds / 750	ms / image
Latent generation	42.9	57.2
VAE decode	40.8	54.4
Generation + decode	83.7	111.7
Full evaluation packet	97.3	129.7

The ablations support the theory in the form stated above. The No-DWT row is the largest geometric drop, which matches the prediction from Theorem 1. The routing rows justify $p = 0.8$: at $p = 0$ or $p = 0.5$, the high-frequency inference path is under-trained, while $p = 1.0$ slightly raises CLIP-S but reopens LPIPS. Endpoint WCT is also necessary: per-step AdaIN raises LPIPS by 0.0975, consistent with Proposition 3. LL weights above the selected setting improve LPIPS slightly but lower CLIP-S, which is why the deployed model keeps LL weak rather than zeroing it out or matching it strongly.

5.4 Controls and Extensions

Table 4 collects the checks that matter beyond the main operating point.

This section tests whether the main story survives outside the exact selected row. First, identity pairs are not carrying the result. Removing them changes the selected WD-VF point only mildly, from 0.7213 / 0.2868 on all pairs to 0.6908 / 0.2965 on the off-diagonal subset. Second, the latent prior matters independently of wavelet routing. On the matched 256×256 control, direct RGB flow reaches 0.6960 / 0.5317,

Table 3: Geometric component ablation. Each row removes, replaces, or perturbs one WD-VF design choice. $\Delta\text{CLIP-S}$ and ΔLPIPS are relative to the WD-VF baseline (0.7213 / 0.2868).

Variant	CLIP-S	LPIPS	$\Delta\text{CLIP-S}$	Note
WD-VF (full)	0.7213	0.2868	—	baseline
<i>Wavelet basis</i>				
No DWT (Euclidean flow)	0.7117	0.2994	-0.0096	content-style conflict
Daubechies-2 (vs. Haar)	0.7258	0.3288	+0.0045	basis choice secondary
<i>Base de-weighting and LL handling</i>				
Lock LL at inference	0.7174	0.3372	-0.0039	removes useful LL drift
$w_{\text{LL}} = 0.5$ at train	0.7174	0.2774	-0.0039	more LL pressure
$w_{\text{LL}} = 1.0$ at train	0.7180	0.2764	-0.0033	stronger LL pressure
Train HH head	0.7213	0.2868	± 0.0001	negligible gain
<i>Endpoint alignment</i>				
Per-step AdaIN	0.7361	0.3843	+0.0148	repeated over-stylization
AdaIN (diagonal) vs. WCT	0.7266	0.3402	+0.0053	weaker cross-channel match
Inject LL too	0.7215	0.3856	+0.0002	global color drift
<i>Style-query routing</i>				
$p = 0.0$ (Euclidean train)	0.7061	0.2606	-0.0152	train-test mismatch
$p = 0.5$	0.7083	0.2480	-0.0130	under-uses high-frequency query path
$p = 1.0$ (always DWT)	0.7226	0.3068	+0.0013	slightly over-constrained
<i>Solver</i>				
Euler (vs. Heun)	0.7210	0.3129	-0.0003	lower solver accuracy
RK4 (vs. Heun)	0.7265	0.3235	+0.0052	extra cost, no gain
<i>Capacity</i>				
depth=2 (vs. 4)	0.7172	0.2705	-0.0041	lower capacity
dim=32 (vs. 64)	0.7153	0.2705	-0.0060	lower width
gate=0 (no gate)	0.7080	0.2641	-0.0133	removes style control

whereas the latent-space control reaches 0.7168 / 0.3125. Third, the transport geometry is reusable. Updating only the style-memory side keeps the system near the main operating point, and the 8-style extension remains viable without full retraining.

Haar is not the only workable wavelet basis, but it is the most reliable default in the current packet. Its exact orthogonality makes Theorem 2 literal, keeps the subband supervision easy to interpret, and gives the better CLIP-S/LPIPS

Table 4: Controls and extension checks. These rows test whether the main claim survives changes in scope, representation, and style set.

Setting	Scope	CLIP-S	LPIPS
WD-VF selected	all-pairs	0.7213	0.2868
WD-VF selected	transfer-only	0.6908	0.2965
Pixel256 control	all-pairs	0.6960	0.5317
Latent256 control	all-pairs	0.7168	0.3125
Style-memory-only up-date	all-pairs	0.7218	0.3020
8-style extension	transfer-only	0.7039	0.2555

tradeoff overall once the Daubechies-2 LPIPS penalty is counted.

5.5 Theory-to-Experiment Alignment

Table 5 links each formal statement to a concrete experimental consequence. The pattern is consistent: removing the wavelet coordinate change hurts the frontier, mismatching train-time and test-time routing costs CLIP-S, repeated per-step injection overshoots LPIPS exactly as the residual-decay argument predicts, and the matched pixel-space control shows that the gain is not a generic consequence of adding another loss term.

6 Discussion

Training efficiency. WD-VF trains in minutes because the routed objective is easier, not because the network is merely small. LL carries most global structure. Once LL is weakly supervised and excluded from the style-query path at inference, the learned transport stops paying to reconcile incompatible low-frequency objectives and spends capacity on high-frequency texture motion instead.

Mechanism of real style transfer. WD-VF edits the high-frequency subbands directly and aligns their endpoint statistics once, at the last step. It does not repeatedly inject style through the whole trajectory. In practice this produces visible stroke and surface-texture transfer while keeping the source layout stable.

Hard targets. Minimalism remains the hardest target family because its style change is driven more by low-frequency palette than by brushwork. This behavior matches the design: LL is weakly supervised during transport and fixed during endpoint stylization. WD-VF therefore prioritizes structure stability over large palette shifts.

Benchmark regime. The present benchmark targets domain-level transfer across the five styles with the lowest IDT CLIP-S from a broader WikiArt pool. The regime is therefore deliberately hostile to no-op behavior. Within it, the decisive choices are the latent semantic prior and the wavelet separation that prevents that prior from collapsing back to source reconstruction.

Table 5: Prediction-to-evidence map for the main theoretical claims. Each row states the predicted behavior and the closest direct observation.

Claim and predicted consequence	Matched evidence
Euclidean latents admit a style-suppressed regime, so removing the wavelet change should weaken the frontier (Theorem 1)	Removing the wavelet change drops WD-VF from 0.7213 / 0.2868 to 0.7117 / 0.2994, showing that the Euclidean parameterization gives a weaker tradeoff even before any large architectural change.
If the high-frequency route is the deployed style path, then train-time route matching should matter (Theorem 2)	Routing with $p = 0.8$ beats $p = 0.0$ and $p = 0.5$ on CLIP-S while staying in the same damage regime, which is exactly the predicted benefit of training the same high-frequency query path used at inference.
If endpoint injection already matches the target subband statistics, repeated injection should overshoot damage (Proposition 3)	Per-step AdaIN raises LPIPS from 0.2868 to 0.3843 despite a higher CLIP-S, matching the predicted saturation of repeated residual shrinkage.
If the latent prior already organizes semantic structure, then pixel-space transport should be markedly less stable (Proposition 1)	On the matched 256×256 control, pixel-space flow reaches 0.6960 / 0.5317 while latent-space flow reaches 0.7168 / 0.3125, which isolates the effect of the latent semantic prior.

Limitations. WD-VF is domain-conditional rather than exemplar-conditional, so it is not intended to copy a specific reference painting. The current benchmark is also narrow by design: it isolates five hard WikiArt styles chosen by low IDT CLIP-S, but it does not yet cover photo-to-art transfer, larger style vocabularies, or broader cross-domain evaluation in the main paper. The LL design makes palette-driven targets such as Minimalism harder than heavily textured targets because the method deliberately protects low-frequency structure. The theory is local and mechanism-oriented: it explains why the deployed coordinate change and routed objective behave well, and it predicts specific ablation patterns, but it does not claim a global convergence theorem for every wavelet flow. Finally, the main table is single-seed. This is acceptable for the present draft because the effect sizes are large and the protocol is fully specified, but multi-seed confirmation would strengthen the final submission.

7 Conclusion

Art-to-art style transfer needs two things at once: the right success criterion and the right coordinate system. IDT calibration exposes when a method is only reconstructing the source. WD-VF escapes that regime by separating low-

frequency structure from high-frequency style motion and by routing style through the same high-frequency path used at inference.

The resulting operating point is unusually affordable: 903K parameters, 3.08 minutes of training on a single RTX 3060, 83.7 seconds to generate and decode 750 outputs, and CLIP-S/LPIPS of 0.7213/0.2868 on Distinct5-WikiArt. The ablations show that the gain is tied to the wavelet change, routed supervision, and endpoint alignment rather than to a generic increase in model complexity. Affordable real style transfer therefore comes from the right geometry, not a large training budget.

Ethics Statement

Style transfer tools can be misused to produce deepfake art or to misattribute stylistic authorship. Our method operates at the domain level (e.g., “Impressionism”) rather than the per-artist level, which reduces the risk of impersonating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt, which consists of public-domain and historically significant artworks; no personal data is used. We encourage downstream users to disclose when an image has been style-transferred and to respect the licensing terms of the source content.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The training configuration, hyperparameters, and evaluation protocol are described in Section 5. Training takes 3.08 minutes on a single RTX 3060 (12 GB). The 750-pair packet needs 83.7 seconds for generation plus VAE decode and 97.3 seconds including CLIP/LPIPS metrics. The 903K-parameter checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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AAAI Reproducibility Checklist

1. This paper claims the following contributions: (1) IDT-calibrated evaluation for art-to-art transfer (Section 5); (2) analysis of Euclidean collapse through the set $\mathcal{M}_0$ and Theorem 1 (Section 3.2); (3) the WD-VF method — wavelet-decoupled rectified flow with high-frequency query routing and endpoint subband alignment (Section 3); (4) objective decomposition in the high-frequency-only query case and its optimization consequence (Section 4).
2. The dataset (Distinct5-WikiArt-512) is publicly available; the evaluation protocol (CLIP-S, LPIPS, 750 pairs including 150 identity pairs) is described in Section 5.
3. Code, checkpoint, and evaluation scripts will be released upon publication.
4. Training requires a single 12 GB GPU (RTX 3060); 5 epochs; 3.08 minutes end-to-end.
5. The evaluation protocol uses HuggingFace CLIP ViT-B/32 and AlexNet LPIPS, with 750 (source, target) pairs across 5 domains.

6. All baseline numbers (Table 1) are evaluated under the same protocol; SaMST and SaMam have explicit convergence packets, while CUT uses a retained Related_Works checkpoint with recorded training time but no surviving training log.
7. Statistical reporting: Table 1 reports the main single-seed 750-pair results. The supplement provides ablations, boundary-case analyses, and extension studies.